

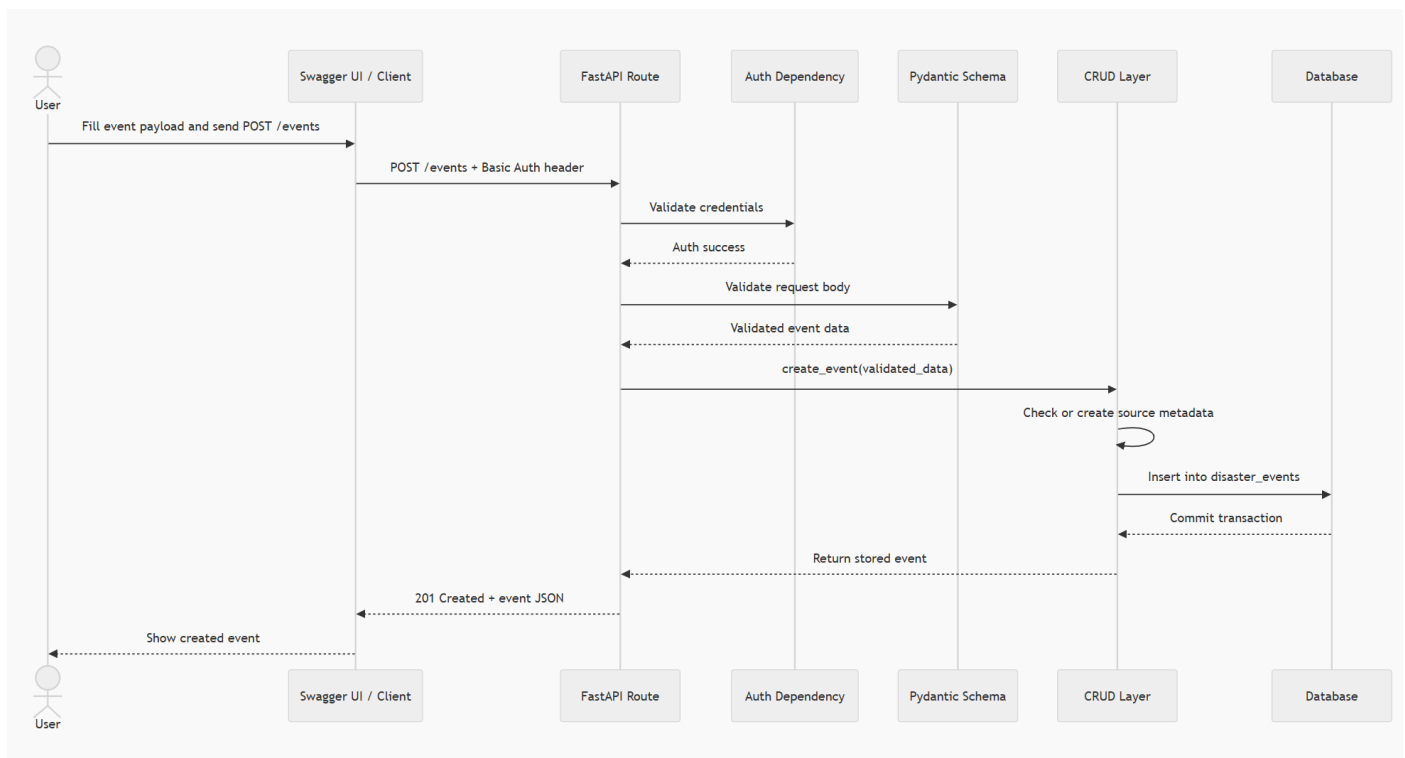
Sequences Diagram

1. Manual Event Creation Sequence

Actor: User

Participants: Swagger UI / Client, FastAPI Route, Auth Dependency, Pydantic Schema, CRUD Layer, Database

1. User sends POST /events with event payload
2. Swagger UI / Client attaches Basic Authentication header
3. FastAPI Route receives the request
4. Auth Dependency validates username and password
5. Pydantic Schema validates request body fields
6. Route calls CRUD Layer to create the event
7. CRUD Layer checks or creates source metadata
8. CRUD Layer inserts the event into disaster_events
9. Database commits the transaction
10. CRUD Layer returns the stored event
11. FastAPI Route returns 201 Created response to the user

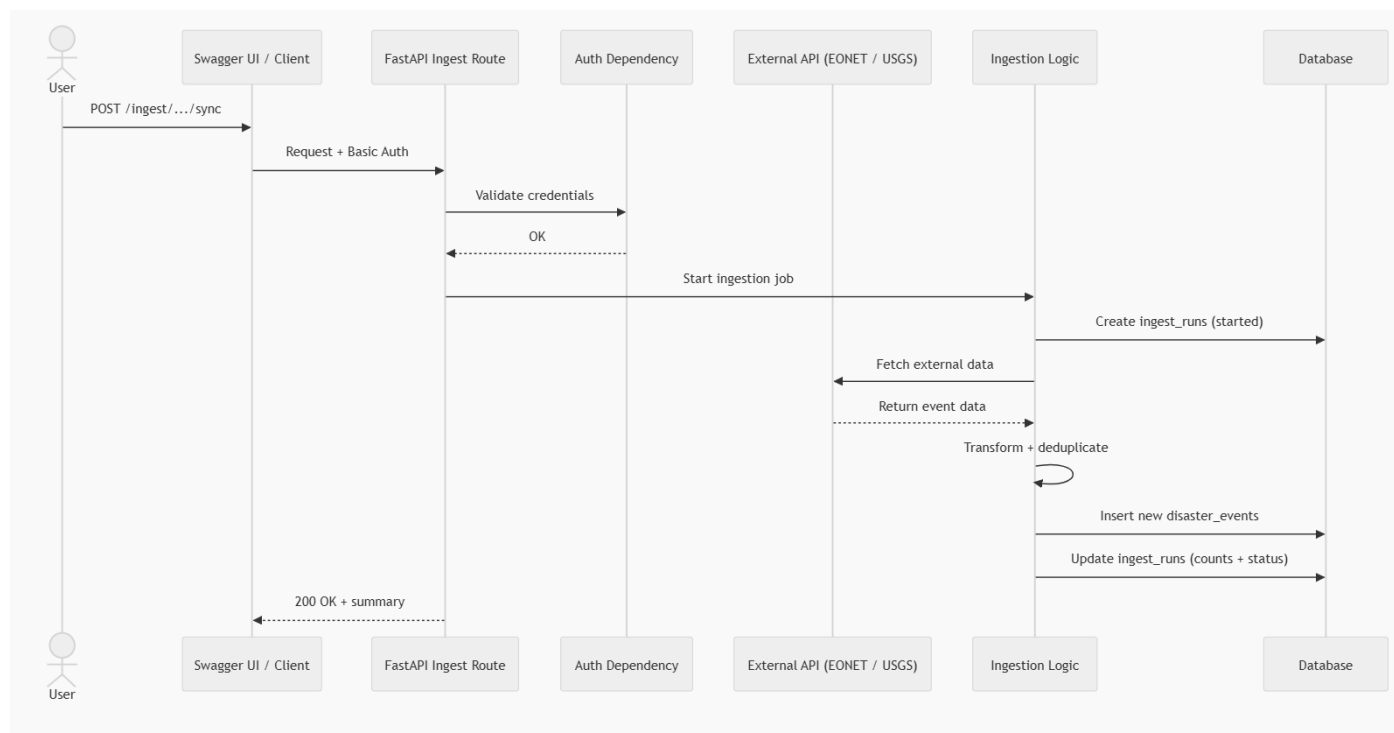


2. External Ingestion Sequence

Actor: User

Participants: Swagger UI / Client, FastAPI Ingest Route, Auth Dependency, External API, CRUD Layer, Database

1. User sends POST /ingest/eonet/wildfires/sync
2. Swagger UI / Client attaches Basic Authentication header
3. FastAPI Ingest Route receives the request
4. Auth Dependency validates credentials
5. Ingest Route creates an ingest_runs record with status "started"
6. Ingest Route sends request to NASA EONET API
7. External API returns wildfire event data
8. Ingest Route transforms upstream records into local event format
9. CRUD Layer checks whether each event already exists
10. CRUD Layer creates source metadata if needed
11. CRUD Layer inserts new disaster_events records
12. Ingest Route updates ingest_runs with inserted/skipped/failed counts
13. FastAPI returns a sync summary response to the user

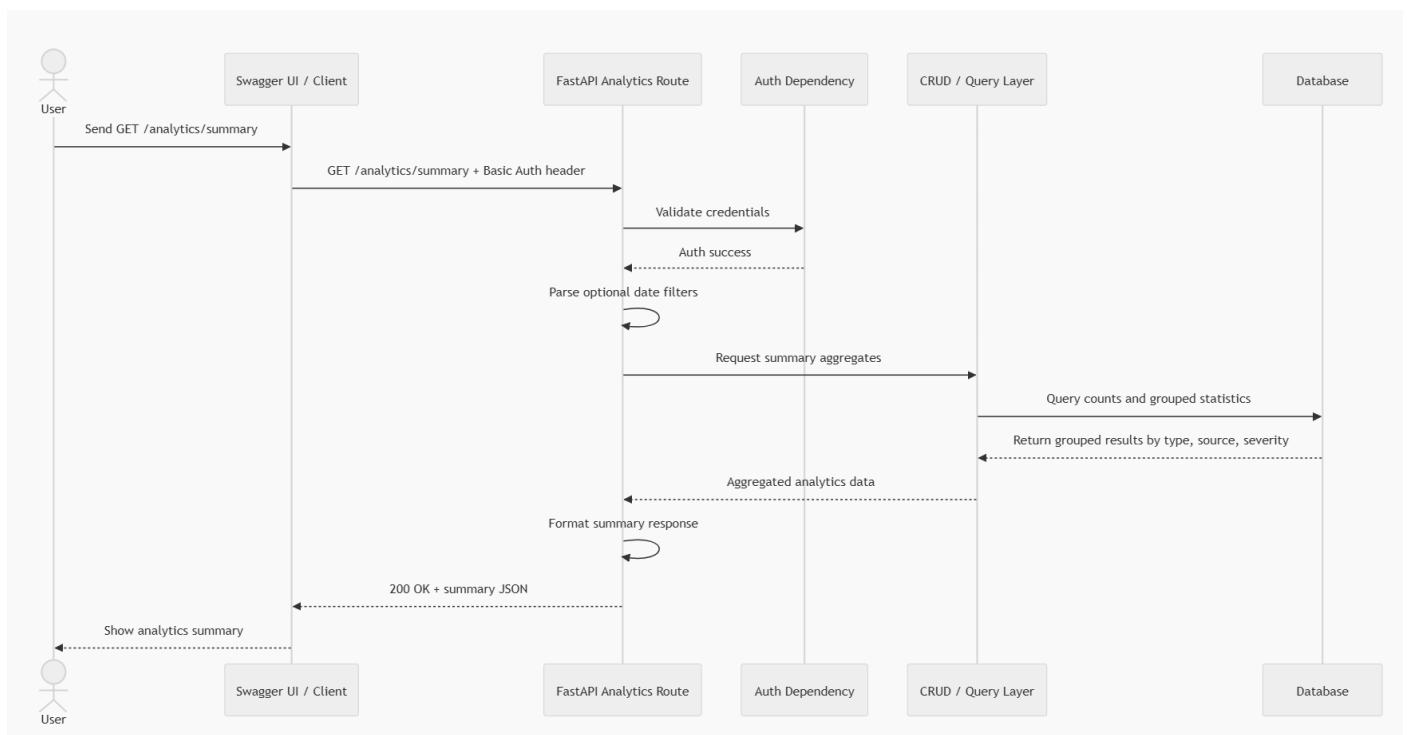


3. Analytics Summary Sequence

Actor: User

Participants: Swagger UI / Client, FastAPI Analytics Route, Auth Dependency, CRUD / Query Layer, Database

1. User sends GET /analytics/summary
2. Swagger UI / Client attaches Basic Authentication header
3. FastAPI Analytics Route receives the request
4. Auth Dependency validates credentials
5. Route parses optional date filters
6. Route queries the local database for event counts and aggregates
7. Database returns grouped statistics by type, source, and severity
8. Analytics Route formats the response
9. FastAPI returns 200 OK with summary analytics data



4. Risk Assessment Sequency

Actor: User

Participants: Swagger UI / Client, FastAPI Analytics Route, Auth Dependency, Database, Risk Scoring Logic

1. User sends GET /analytics/risk-assessment
2. Swagger UI / Client attaches Basic Authentication header
3. FastAPI Route receives the request
4. Auth Dependency validates credentials
5. Route queries recent disaster events from the database
6. Database returns matching events
7. Risk Scoring Logic calculates risk score for each event using severity, type, recency, and local density
8. Route sorts events by risk score
9. FastAPI returns ranked risk assessment results

